

Advanced Embedded Systems Lab

Lab06: Raspberry Pi

Hochschule Hamm-Lippstadt
Summer Term 2025
02.06.2026

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Introduction

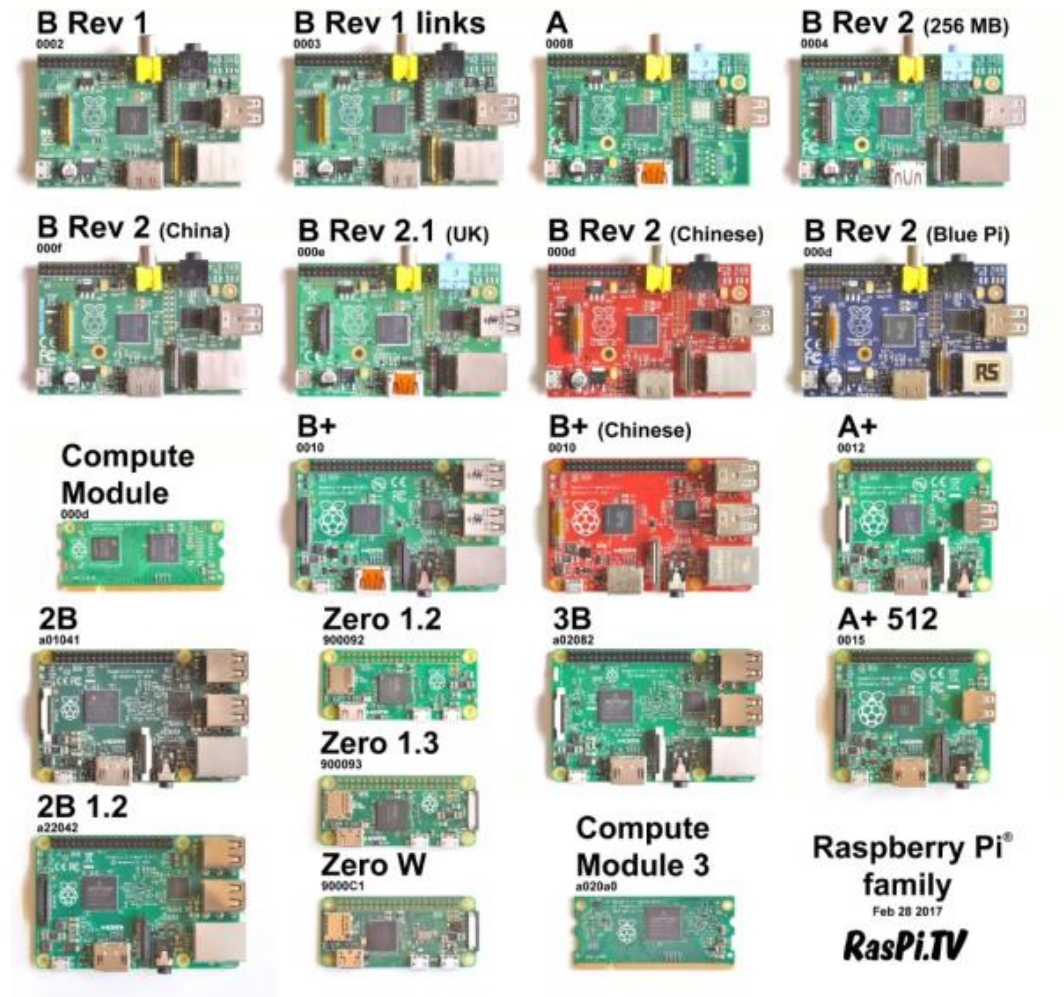
- What is the path that the electrical signal from a sensor takes to end up in a report?
 - Requirements,
 - Use case Diagrams,
 - Block diagrams,
 - sequence diagrams, etc.
- Document your implementation, follow standards, take pictures and videos, etc.
- The more you implement for visualization on Pi-side the better

Raspberry Pi 3 Model B

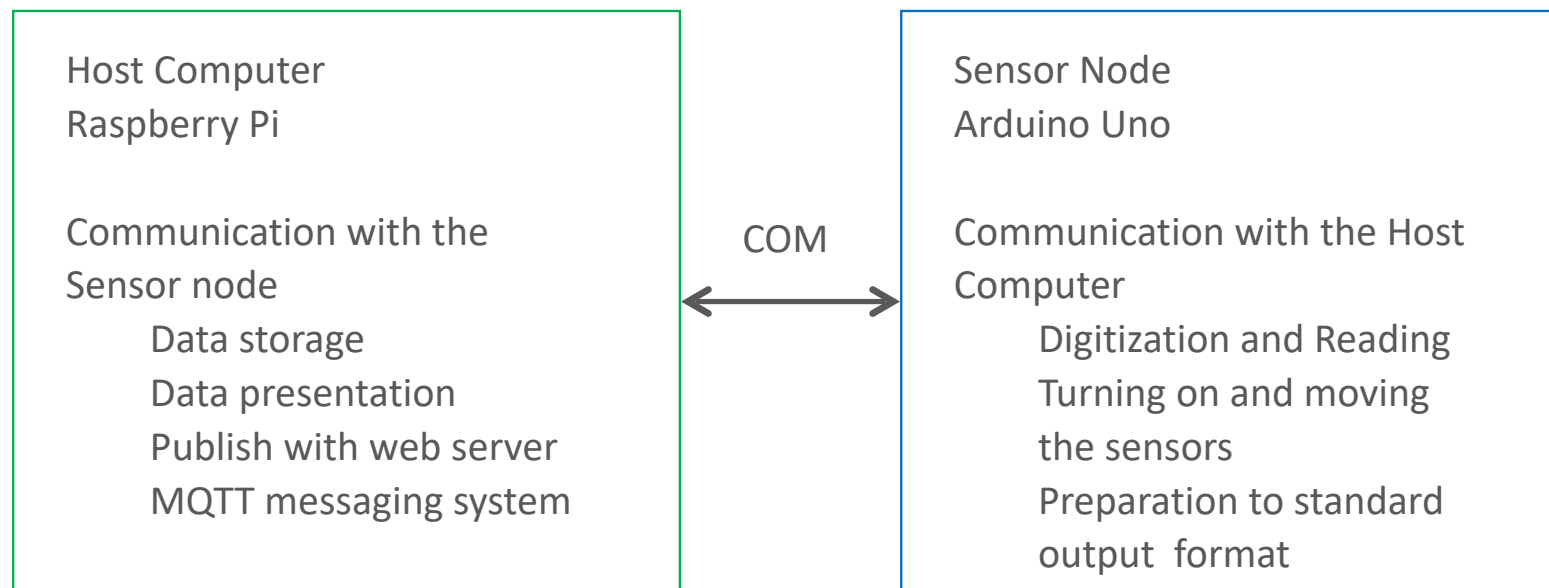
- Raspberry Pi 3 Model B specifications:
 - SoC: Broadcom BCM2837.
 - CPU: 4× ARM Cortex-A53, 1.2GHz.
 - GPU: Broadcom VideoCore IV.
 - RAM: 1GB LPDDR2 (900 MHz)
 - Networking: 10/100 Ethernet, 2.4GHz 802.11n wireless.
 - Bluetooth: Bluetooth 4.1 Classic, Bluetooth Low Energy.
 - Storage: microSD.
 - GPIO: 40-pin header.
 - Ports: HDMI, 3.5mm analogue audio-video jack, 4× USB 2.0, Ethernet, Camera Serial Interface (CSI), Display Serial Interface (DSI).



Raspberry Pi Family



Introduction



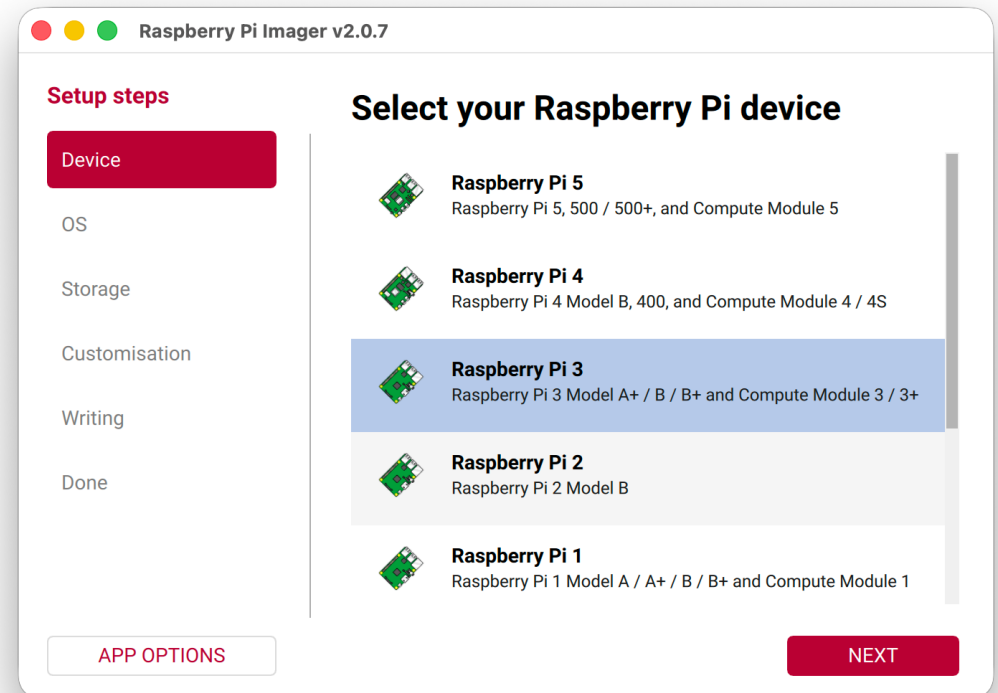
Raspberry Pi

- Safe Handling the Raspberry Pi:
 - Raspberry Pi is a circuit board, not protected, fragile.
 - Ground yourself to remove static electricity (touch a metal surface).
 - Don't place the Raspberry Pi on a metal surfaces.
 - Be careful when connecting and disconnecting power and USB connectors.
 - Be careful not to short out the GPIO pins.
 - A case is a good option.
 - **Re-flashing the SD Cards should be the last option**



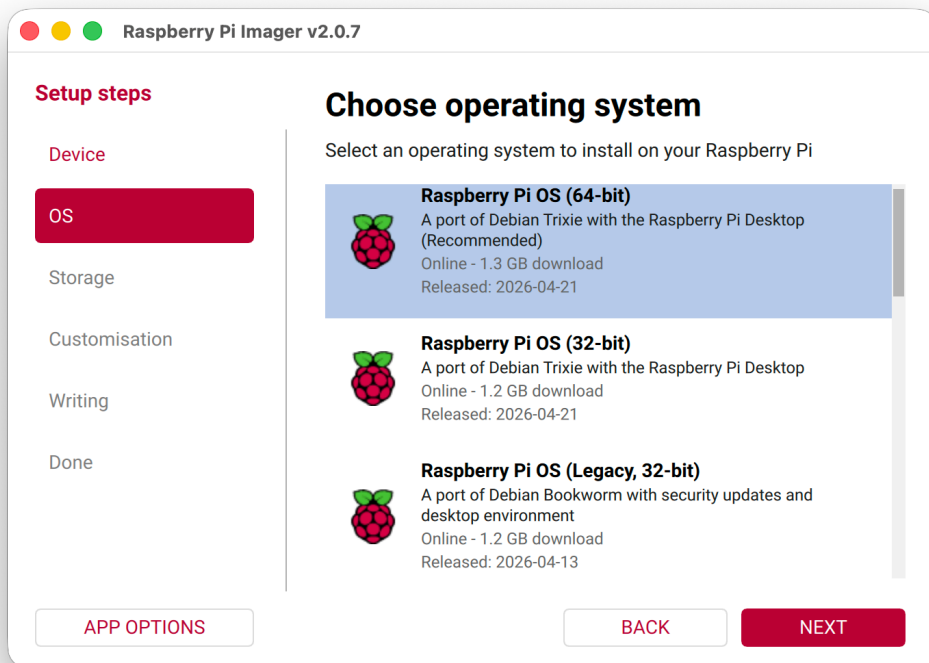
Setting up

- Visit: <https://www.raspberrypi.com/software/>
- Download and install the **Raspberry Pi Imager** which is a quick and easy way to install Raspberry Pi OS
- Plug in your SD card to your computer.
- Launch raspberry pi Imager
- Select your board

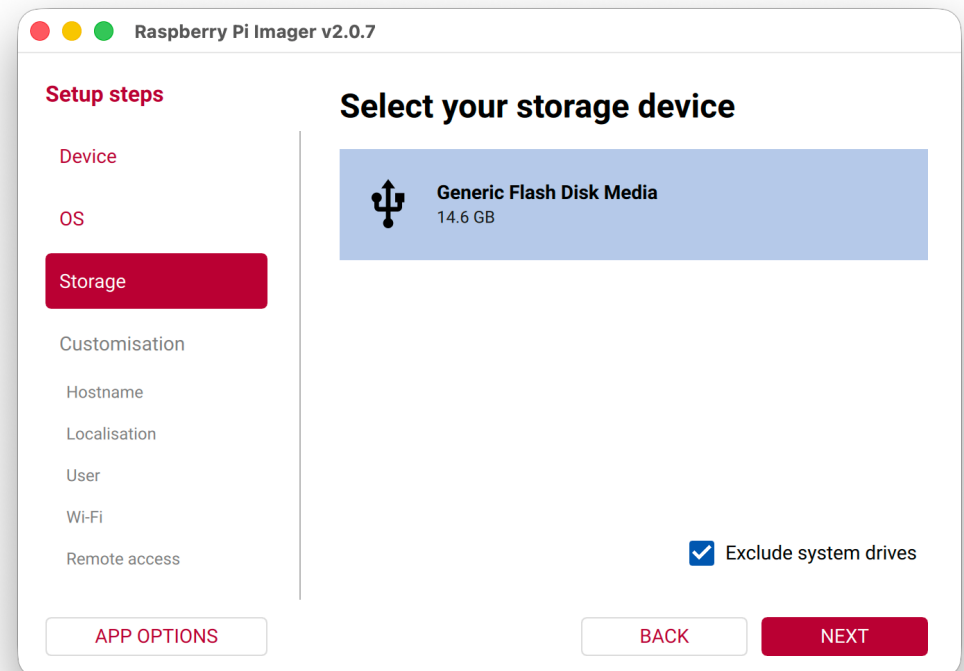


Setting up

- Select your operating system

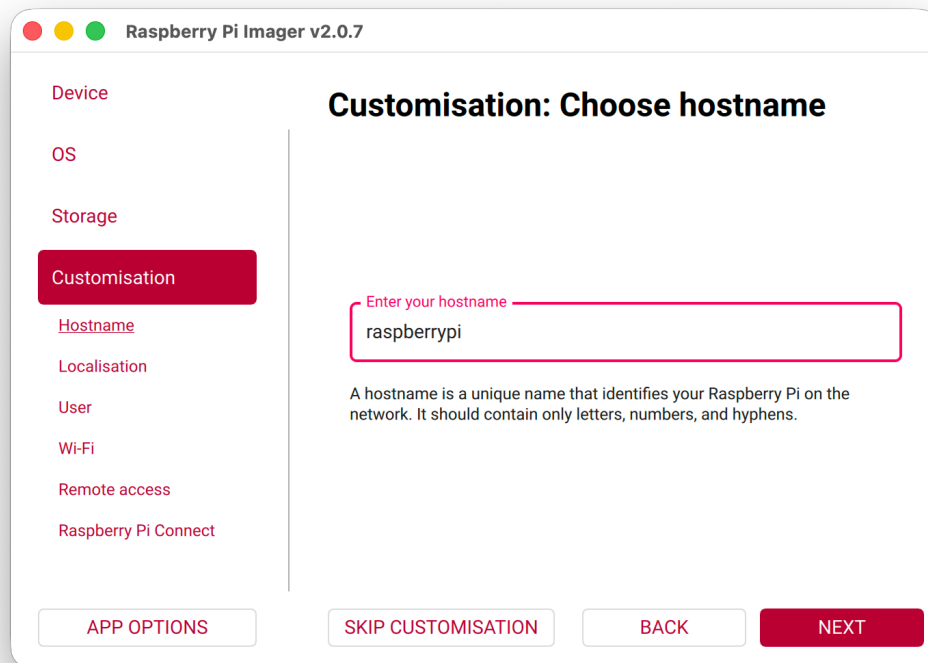


- then your SD card



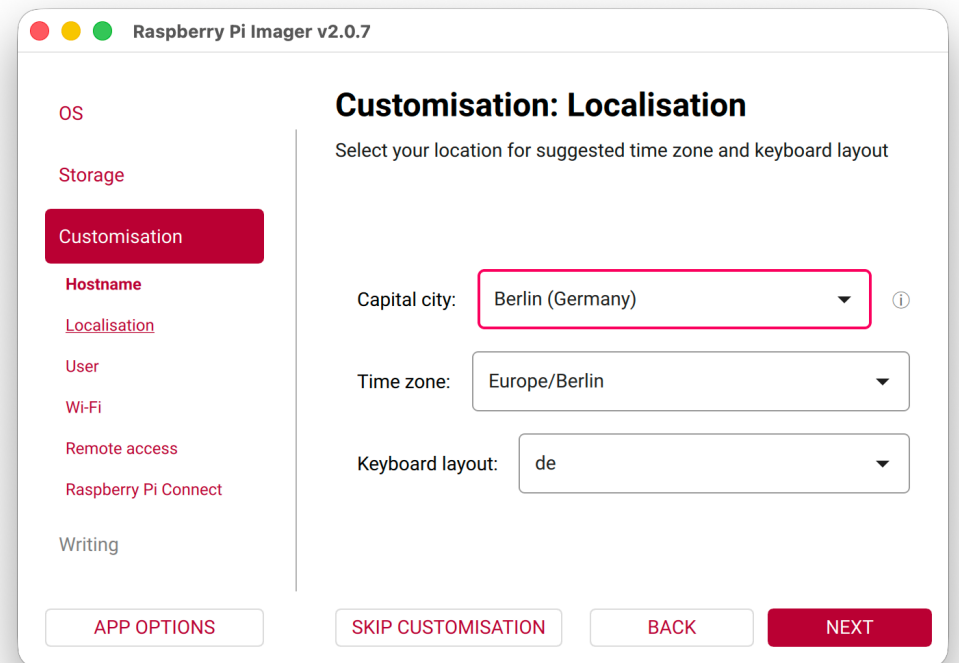
Setting up

- Set your hostname (raspberrypiA#)



The screenshot shows the 'Customisation: Choose hostname' screen in the Raspberry Pi Imager v2.0.7 application. On the left, a sidebar lists various settings: Device, OS, Storage, Customisation (highlighted in red), Hostname, Localisation, User, Wi-Fi, Remote access, and Raspberry Pi Connect. The main area has the title 'Customisation: Choose hostname'. Below it, a text input field is labeled 'Enter your hostname' and contains the text 'raspberrypi'. A small red box highlights the input field. Below the input field, a note states: 'A hostname is a unique name that identifies your Raspberry Pi on the network. It should contain only letters, numbers, and hyphens.' At the bottom, there are four buttons: 'APP OPTIONS', 'SKIP CUSTOMISATION', 'BACK', and 'NEXT' (highlighted in red).

- and your localization



The screenshot shows the 'Customisation: Localisation' screen in the Raspberry Pi Imager v2.0.7 application. On the left, a sidebar lists various settings: OS, Storage, Customisation (highlighted in red), Hostname, Localisation, User, Wi-Fi, Remote access, Raspberry Pi Connect, and Writing. The main area has the title 'Customisation: Localisation' and a subtitle 'Select your location for suggested time zone and keyboard layout'. Below this, there are three dropdown menus: 'Capital city:' with 'Berlin (Germany)' selected, 'Time zone:' with 'Europe/Berlin' selected, and 'Keyboard layout:' with 'de' selected. Each dropdown menu has a small red box highlighting it. At the bottom, there are four buttons: 'APP OPTIONS', 'SKIP CUSTOMISATION', 'BACK', and 'NEXT' (highlighted in red).

Setting up

- Set user name and password

Raspberry Pi Imager v2.0.7

Storage

Customisation

Customisation: Choose username

Create a user account for your Raspberry Pi

Username:

Password:

Confirm password:

The username must be lowercase and contain only letters, numbers, underscores, and hyphens.

APP OPTIONS SKIP CUSTOMISATION BACK NEXT

- and add your hotspot

Raspberry Pi Imager v2.0.7

Storage

Customisation

Customisation: Choose Wi-Fi

SECURE NETWORK **OPEN NETWORK**

SSID:

Password:

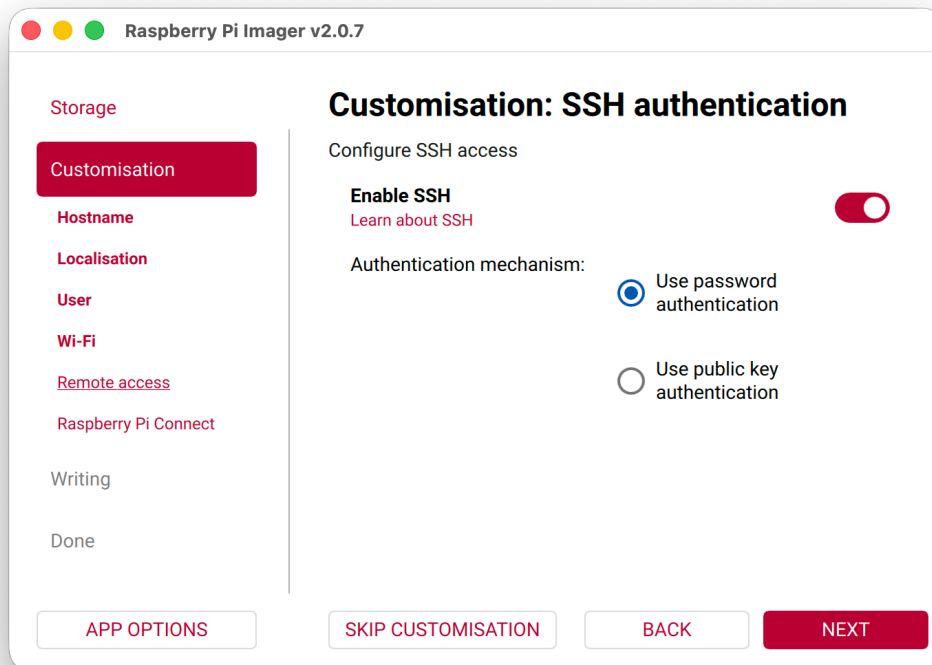
Confirm password:

☐ Hidden SSID

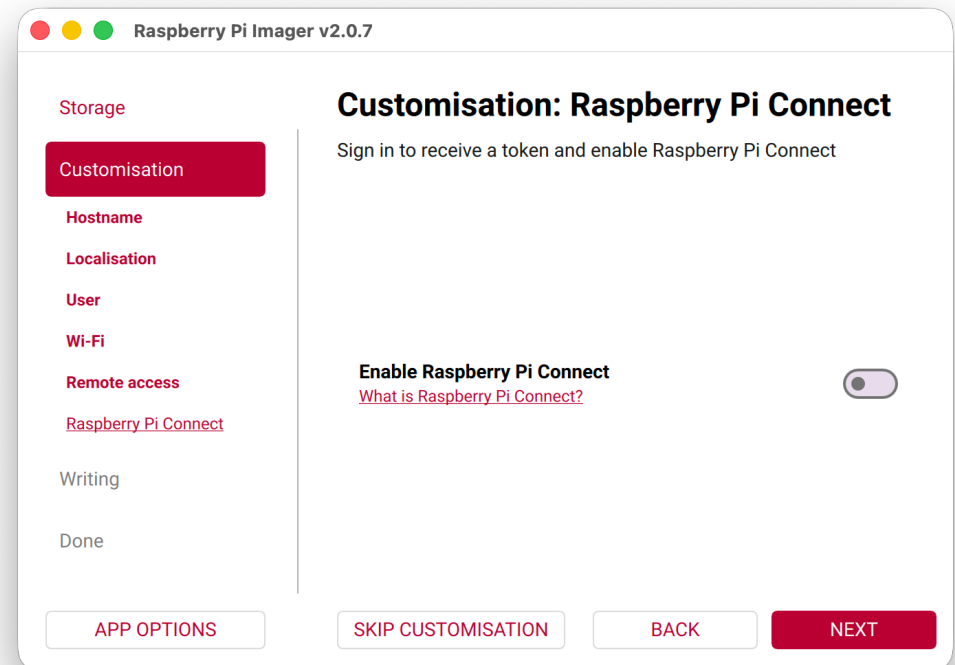
APP OPTIONS SKIP CUSTOMISATION BACK NEXT

Setting up

- Enable SSH with password

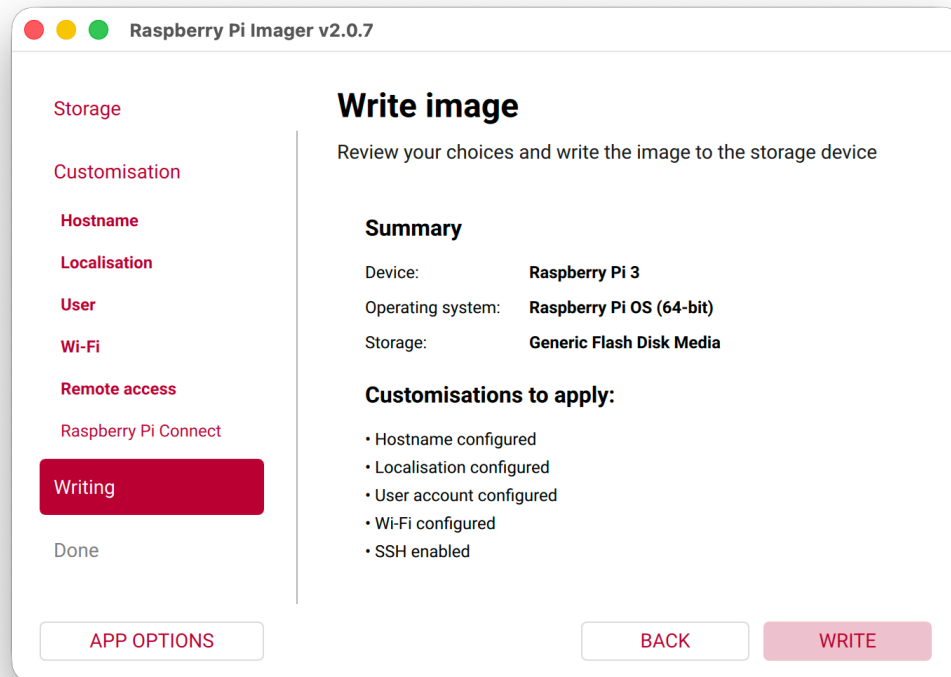


- optionally,

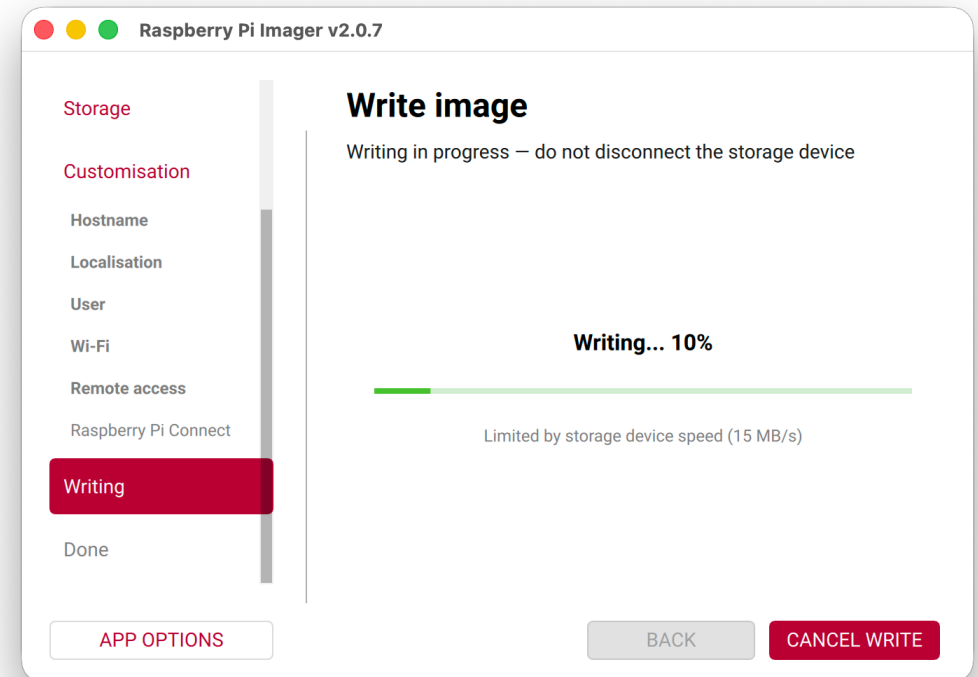


Setting up

- Write to disk,



- and wait

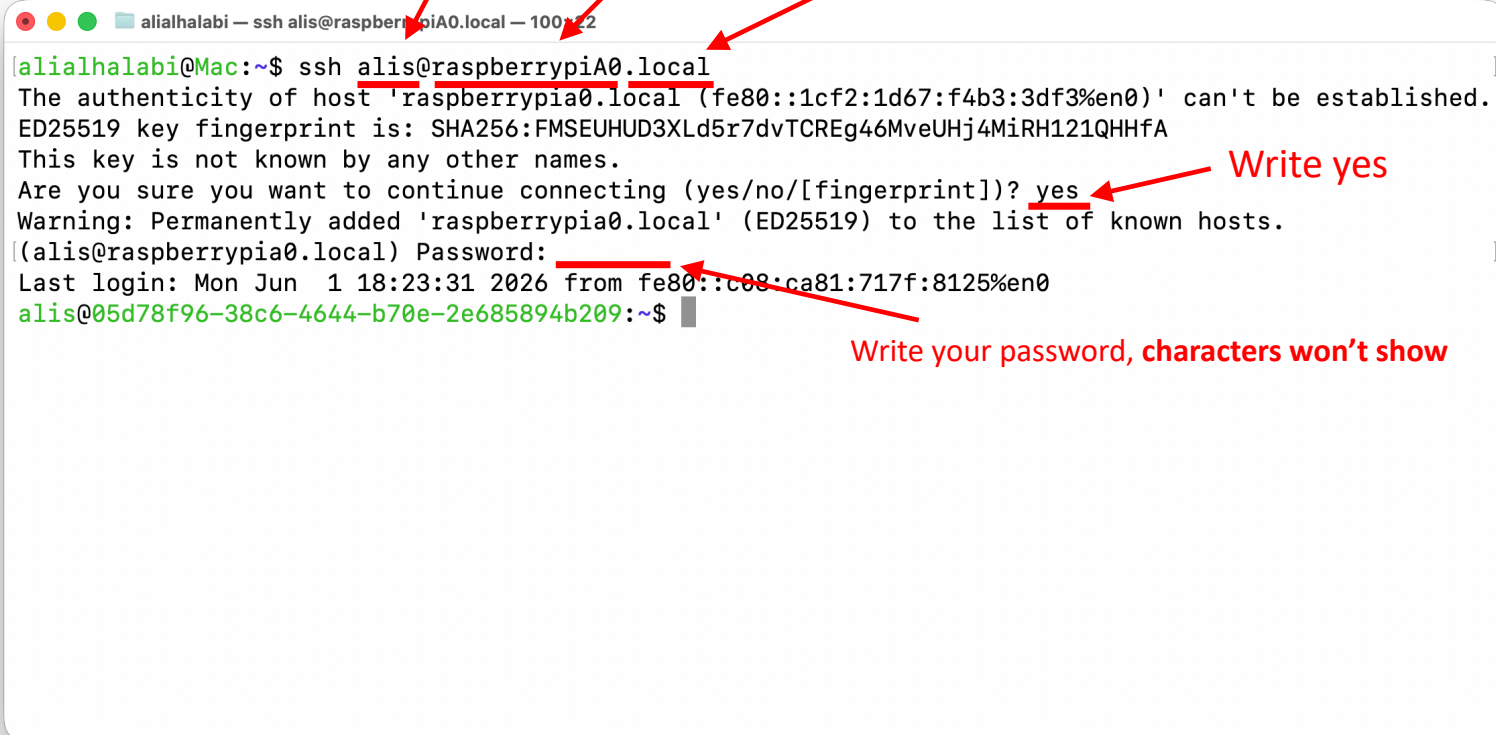


Headless Raspberry Pi

- Insert the flashed SD card to your Pi,
- Connect your Pi to Power (make sure you have grounded yourself, and the Pi is not touching any metals, i.e. your Laptop),
- Make sure your Access point (hotspot or router) is active,
- Wait some minutes.

Headless Raspberry Pi

- With a terminal,
SSH [1] into your Pi using your user name, hostname, and password
(Slide 10 and 11)



The image shows a terminal window titled "alialhalabi — ssh alis@raspberrypiA0.local — 100x122". The terminal output is as follows:

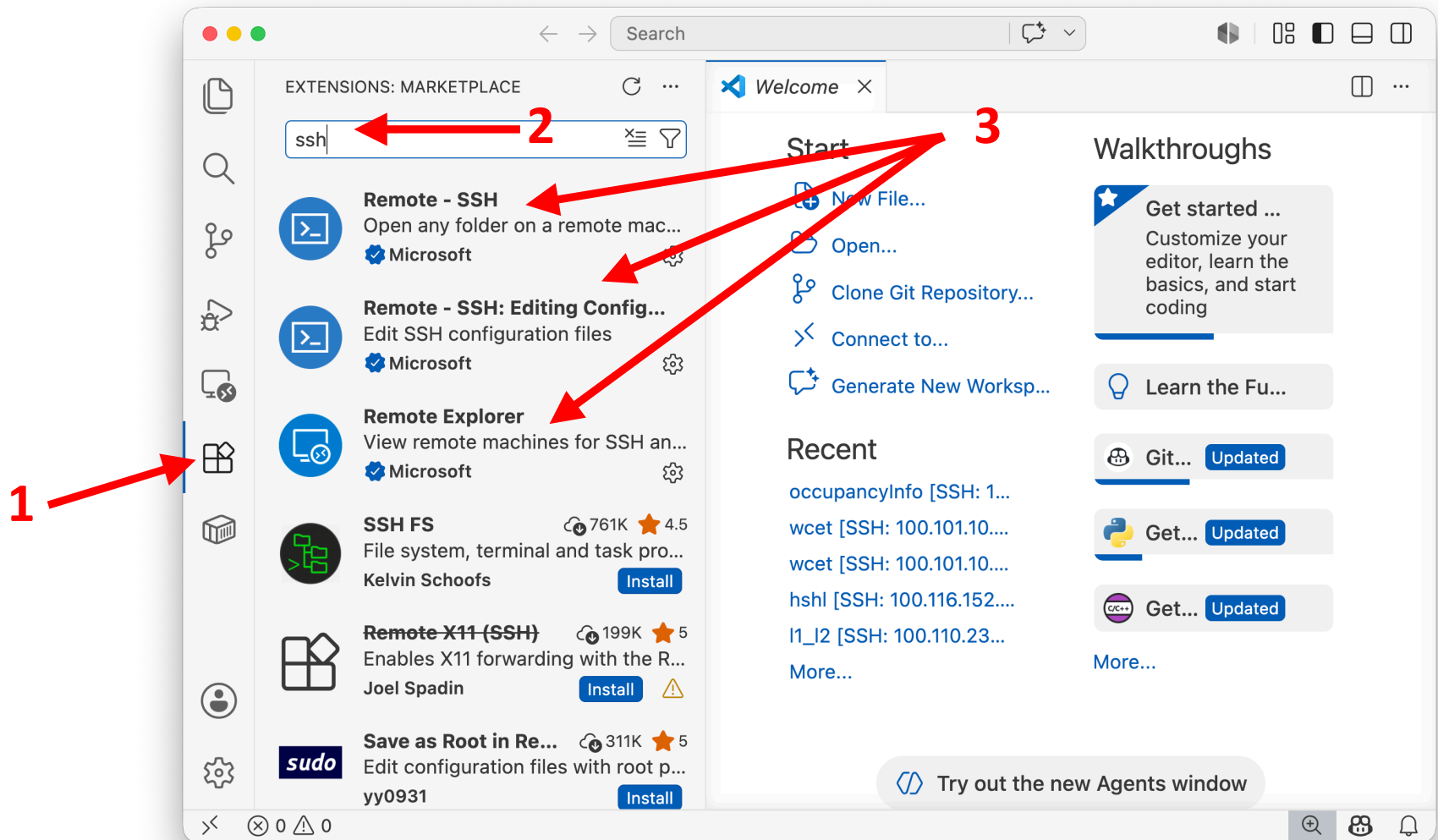
```
[alialhalabi@Mac:~]$ ssh alis@raspberrypiA0.local
The authenticity of host 'raspberrypiA0.local (fe80::1cf2:1d67:f4b3:3df3%en0)' can't be established.
ED25519 key fingerprint is: SHA256:FMSEUHUD3XLd5r7dvTCREg46MveUHj4MiRH121QHHfA
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'raspberrypiA0.local' (ED25519) to the list of known hosts.
(alis@raspberrypiA0.local) Password:
Last login: Mon Jun  1 18:23:31 2026 from fe80::c08:ca81:717f:8125%en0
alis@05d78f96-38c6-4644-b70e-2e685894b209:~$
```

Red arrows point to specific parts of the terminal output with the following annotations:

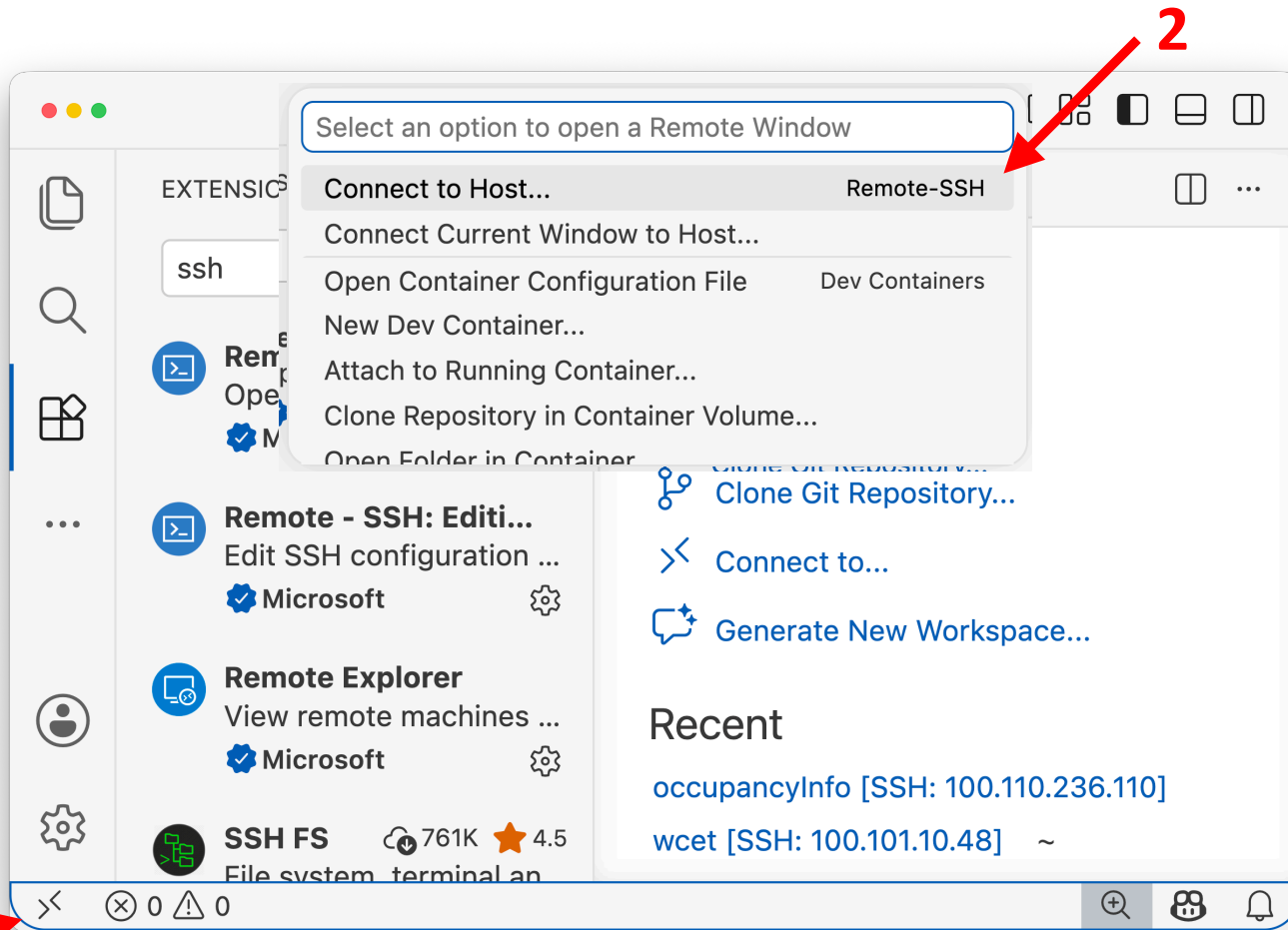
- An arrow points to the command `ssh alis@raspberrypiA0.local` with the text "Only from MacOs".
- An arrow points to the prompt `Are you sure you want to continue connecting (yes/no/[fingerprint])? yes` with the text "Write yes".
- An arrow points to the password prompt `Password:` with the text "Write your password, characters won't show".

[1] <https://www.openssh.org/>

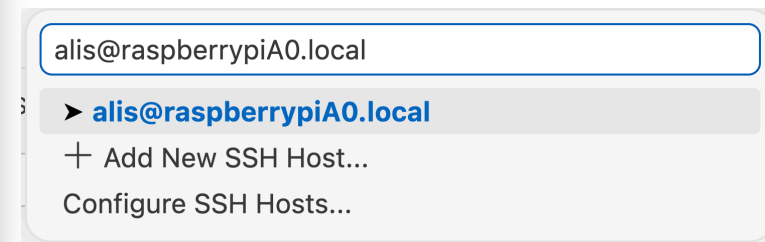
With VSCode



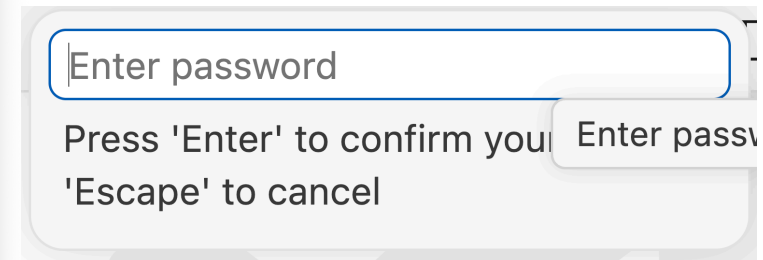
With VSCode



3



4



With Other options

- An HDMI display, keyboard, and mouse.
- A VNC server.

Important **BASH** commands

alialhalabi — ssh alis@raspberrypiA0.local — 48x5

```
alis@raspberrypiA0:~$ sudo apt update
```

alialhalabi — ssh alis@raspberrypiA0.local — 48x5

```
alis@raspberrypiA0:~$ sudo apt upgrade
```

General BASH commands

alialhalabi — ssh alis@raspberrypiA0.local — 44x5

```
alis@raspberrypiA0:~$ clear
```

alialhalabi — ssh alis@raspberrypiA0.local — 70x5

```
alis@raspberrypiA0:~$ ls
Applications  Downloads  Library  'Oskar Stalberg'  Qt
Desktop       Games     Movies   Pictures          QucsWorkspace
Documents     LTspice   Music    Public            pihole
alis@raspberrypiA0:~$
```

alialhalabi — ssh alis@raspberrypiA0.local — 47x5

```
alis@raspberrypiA0:~$ mkdir newDir
alis@raspberrypiA0:~$ cd newDir
alis@raspberrypiA0:~/newDir$ cd ..
alis@raspberrypiA0:~$ rmdir newDir
alis@raspberrypiA0:~$
```

alialhalabi — ssh alis@raspberrypiA0.local — 47x9

```
alis@raspberrypiA0:~$ mkdir newDir
alis@raspberrypiA0:~$ cd newDir
alis@raspberrypiA0:~/newDir$ touch newfile
alis@raspberrypiA0:~/newDir$ ls
newfile
alis@raspberrypiA0:~/newDir$ pwd
/Users/alis/newDir
alis@raspberrypiA0:~/newDir$ cd ..
alis@raspberrypiA0:~$ rm -r newDir
```

alialhalabi — ssh alis@raspberrypiA0.local — 41x5

```
alis@raspberrypiA0:~$ touch newFile.txt
alis@raspberrypiA0:~$ nano newFile.txt
```

General **BASH** commands

Navigation

Cmd	Argument	Description
Pwd		Show current directory
ls		List files and folders
ls	-la	Detailed list including hidden files
cd	folder	Change directory
cd	..	Go up one directory
cd	~	Go to home directory

What else?

Files and Directories

Cmd	Argument	Description		
touch	file.txt	Create empty file		
mkdir	myfolder	Create directory		
cp	file1 file2	Copy file		
cp -r	dir1 dir2	Copy directory		
mv	old.txt new.txt	Rename/move file		
rm	file.txt	Delete file		
rm -r	folder	Delete directory		

WLAN Connection

```
$ sudo nano /etc/etc/wpa_supplicant/wpa_supplicant.conf
```

```
ctrl_interface=DIR=/var/run/wpa_supplicant GROUP=netdev  
update_config=1  
network={  
    ssid="Euer_WLAN_Name"  
    psk="WLAN_Password"  
    key_mgmt=WPA-PSK  
}
```

Python Program

Makes more sense to use VSCode ssh

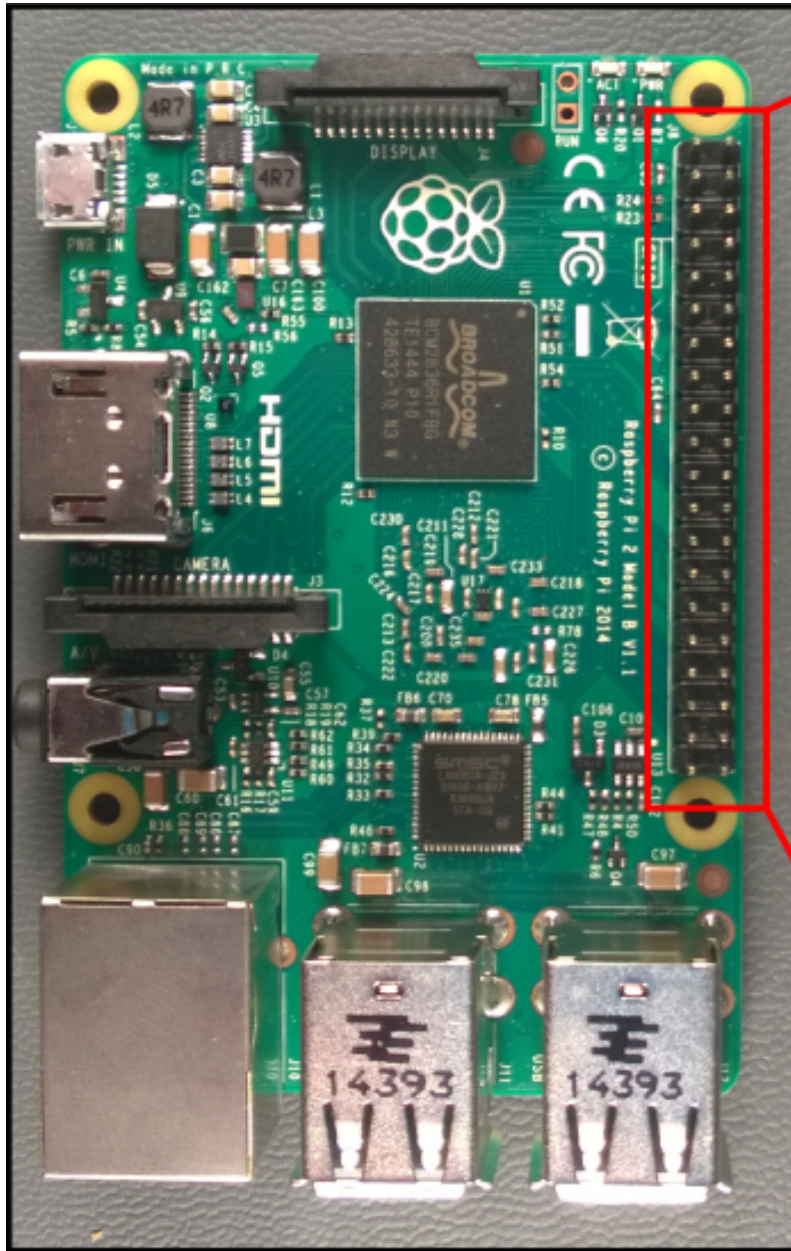
```
alis@raspberrypiA0:~$ touch pyCode.py
alis@raspberrypiA0:~$ nano pyCode.py
```

```
UW PICO 5.09 File: pyCode.py Modified
print("Hello World 🌍")
```

^G Get Help ^O WriteOut ^R Read File ^Y Prev Pg ^K Cut Text ^C Cur Pos
^X Exit ^J Justify ^W Where is ^V Next Pg ^U UnCut Text ^T To Spell

Control + X —> Y —> check file name —> return key

```
alis@raspberrypiA0:~$ python3 pyCode.py
Hello World 🌍
alis@raspberrypiA0:~$
```



3.3V PWR	1	2	5V PWR
I2C1 SDA	3	4	5V PWR
I2C1 SCL	5	6	GND
GPIO 4	7	8	Reserved
GND	9	10	Reserved
SPI1 CS0	11	12	GPIO 18
GPIO 27	13	14	GND
GPIO 22	15	16	GPIO 23
3.3V PWR	17	18	GPIO 24
SPI0 MOSI	19	20	GND
SPI0 MISO	21	22	GPIO 25
SPI0 SCLK	23	24	SPI0 CS0
GND	25	26	SPI0 CS1
Reserved	27	28	Reserved
GPIO 5	29	30	GND
GPIO 6	31	32	GPIO 12
GPIO 13	33	34	GND
SPI1 MISO	35	36	GPIO 16
GPIO 26	37	38	SPI1 MOSI
GND	39	40	SPI1 SCLK

Thank you for your attention!